

## Reposys Campus Reprography Automation System

### *Software Engineering System Design & Proposal Documentation*

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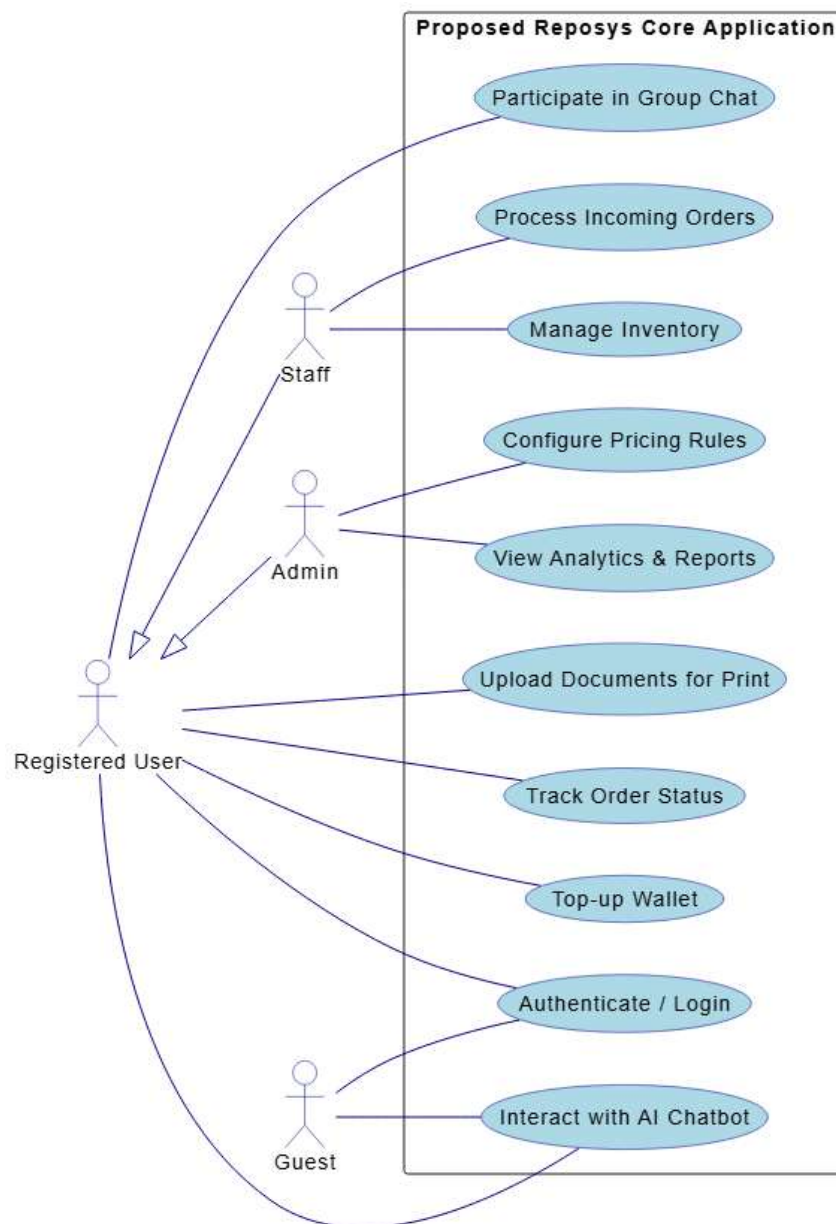
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## 1. System Architecture & UML Diagrams

This section provides a rigorous design blueprint of the proposed system's architecture using standard Unified Modeling Language (UML) notations. These diagrams act as the core structural foundation for the upcoming development phase.

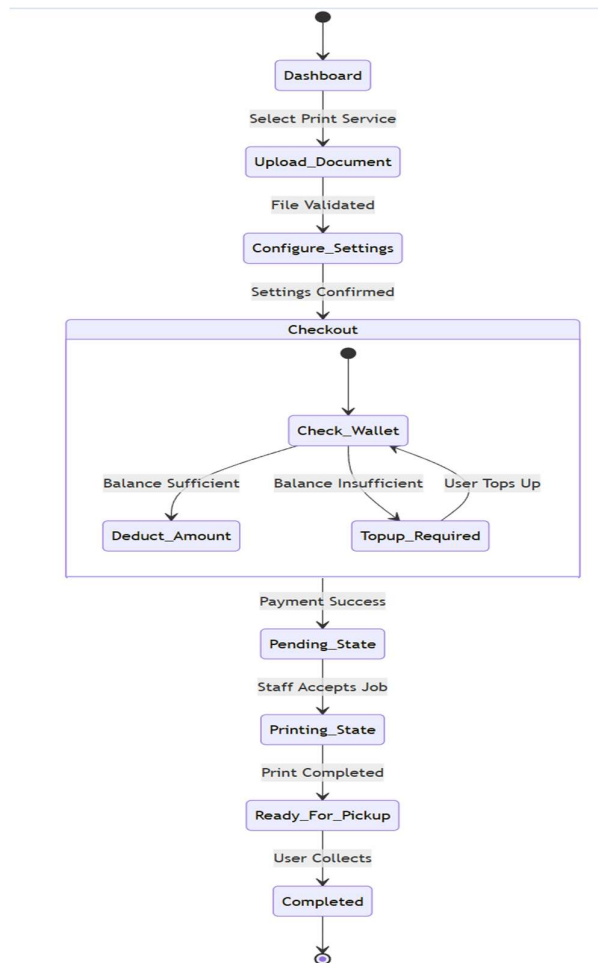
### 1.1 Use Case Diagram

The Use Case diagram identifies the primary actors (Registered Users, Guests, Staff, Admin) and the various system functionalities they will interact with. Students and Teachers share the exact same functional privileges within the system and are collectively modeled simply as "Users".



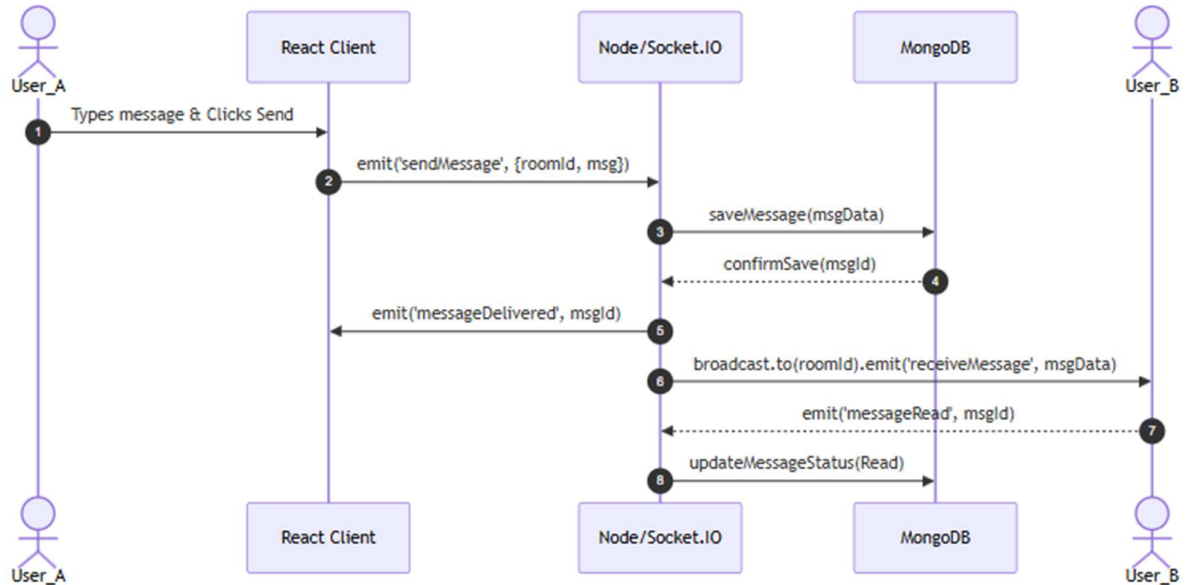
## 1.2 Activity Diagram: Proposed Document Print Workflow

This activity diagram illustrates the intended sequential flow of a student uploading a document, configuring the print settings, completing the digital payment, and tracking the job status.



### 1.3 Sequence Diagram: Real-Time Chat System Architecture

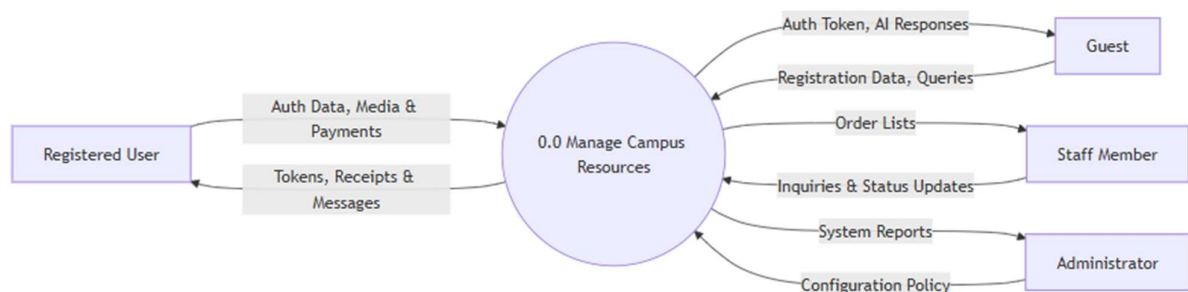
The sequence diagram maps out the proposed bidirectional WebSocket communication for the real-time chat module.



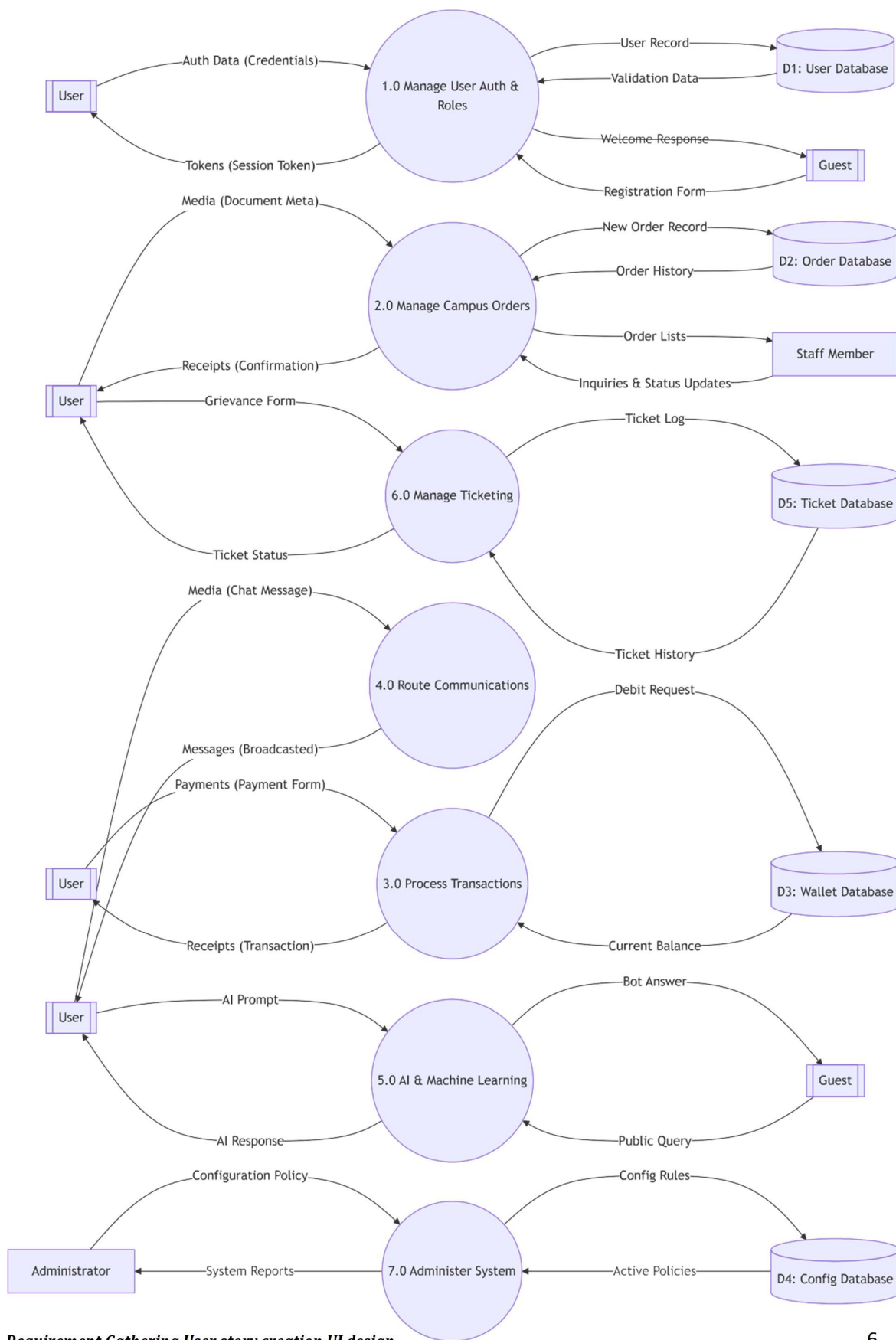
### 1.4 Data Flow Diagrams

#### Level 0 Context Diagram

The Level 0 Context Diagram defines the external entities interacting with the proposed Repsys boundary.

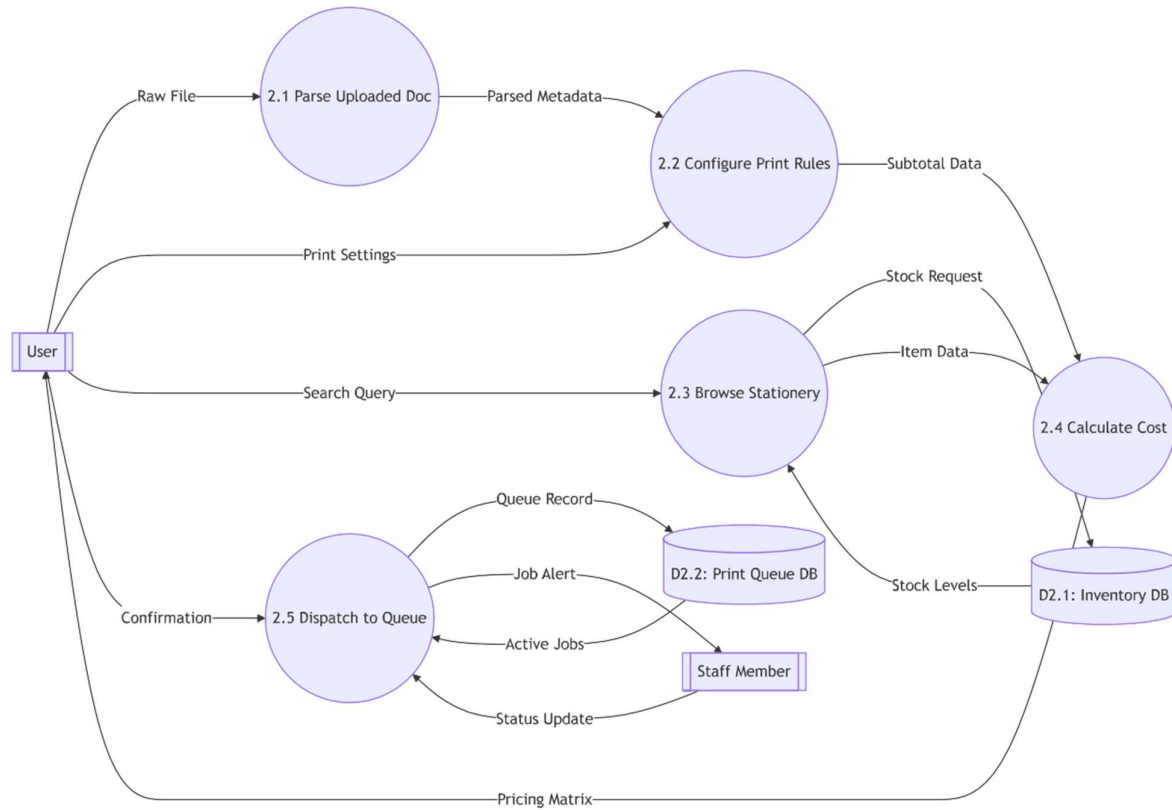


## Level 1 Data Flow Diagram (Main Systems)

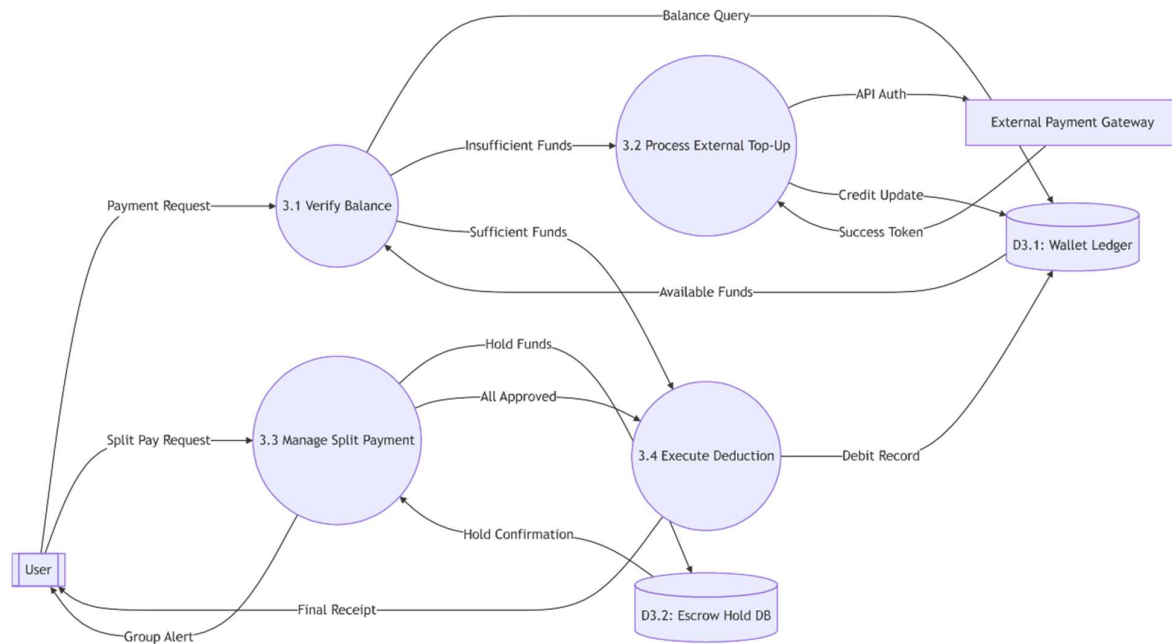


## Level 2 Data Flow Diagram: 2.0 Manage Campus Orders

This Level 2 DFD explodes process 2.0 to demonstrate the granular data flows for Document Parsing, Printing, Stationery Inventory, and Cart management.

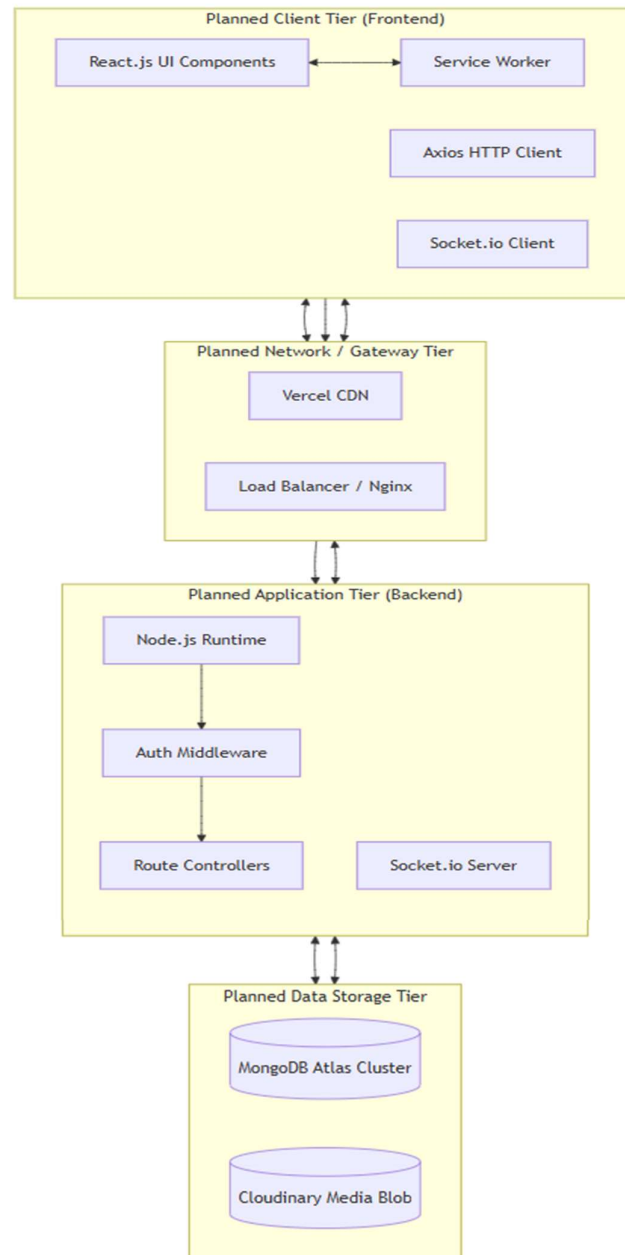


## Level 2 Data Flow Diagram: 3.0 Process Transactions



## 2.5 Component & Deployment Diagram

This diagram represents the planned cloud-native deployment strategy, illustrating how client applications will communicate with the backend services through an API Gateway and load balancer.



## 2. API & Backend Route Definitions

The backend architecture will expose a RESTful API wrapped with custom JWT authentication middleware, strict IP-based rate limiting, and role-based access control models. The following endpoints will form the backbone of client-server data exchange.

### Planned API Architecture Standards

- **Status Codes:** The system will adhere to standard REST HTTP codes: 200 (OK), 201 (Created), 400 (Bad Request), 401 (Unauthorized), 403 (Forbidden), 404 (Not Found), and 500 (Internal Server Error).
- **Response Envelope:** All APIs will be required to return a uniform JSON format: { "success": boolean, "data": Object/Array, "message": string }.

**Middleware Stack:** Every incoming request will systematically pass through Helmet (for Security Headers), CORS origin verifiers, Express JSON parser, Rate Limiter, and finally JWT token verifiers before hitting the controller logic.

| Planned HTTP Method | Endpoint Route (URI)       | Mapped Controller | Auth Requirement       |
|---------------------|----------------------------|-------------------|------------------------|
| GET                 | /api/admin/dashboard-stats | admin.js          | Secured (JWT Required) |
| GET                 | /api/admin/pricing         | admin.js          | Secured (JWT Required) |
| PATCH               | /api/admin/pricing         | admin.js          | Secured (JWT Required) |
| PATCH               | /api/admin/shop/toggle     | admin.js          | Secured (JWT )         |

|               |                                                   |                 |                               |
|---------------|---------------------------------------------------|-----------------|-------------------------------|
| <b>POST</b>   | <b>/api/admin/follow-schedule</b>                 | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>PATCH</b>  | <b>/api/admin/queue/reorder</b>                   | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/admin/activity-logs</b>                   | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>DELETE</b> | <b>/api/admin/activity-logs</b>                   | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/admin/users</b>                           | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/admin/users/create-staff</b>              | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>PATCH</b>  | <b>/api/admin/users/:id/toggle-active</b>         | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>PATCH</b>  | <b>/api/admin/users/:id/limit-override</b>        | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/admin/users/:id</b>                       | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/admin/transactions</b>                    | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/admin/group-orders/:orderId/breakdown</b> | <b>admin.js</b> | <b>Secured (JWT Required)</b> |

|               |                                              |                 |                               |
|---------------|----------------------------------------------|-----------------|-------------------------------|
| <b>GET</b>    | <b>/api/admin/ratings/summary</b>            | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/admin/inquiries</b>                  | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>PATCH</b>  | <b>/api/admin/inquiries/:id/status</b>       | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>PATCH</b>  | <b>/api/admin/inquiries/:id/respond</b>      | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>DELETE</b> | <b>/api/admin/inquiries/:id</b>              | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/admin/staff</b>                      | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>DELETE</b> | <b>/api/admin/staff/:id</b>                  | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>PATCH</b>  | <b>/api/admin/users/:id/approve-role</b>     | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>PATCH</b>  | <b>/api/admin/users/:id/reject-role</b>      | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/admin/staff/assignments</b>          | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>PATCH</b>  | <b>/api/admin/users/:id/queue-assignment</b> | <b>admin.js</b> | <b>Secured (JWT Required)</b> |

|               |                                      |                 |                               |
|---------------|--------------------------------------|-----------------|-------------------------------|
| <b>GET</b>    | <b>/api/admin/users/:id</b>          | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>PUT</b>    | <b>/api/admin/users/:id</b>          | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>DELETE</b> | <b>/api/admin/users/:id</b>          | <b>admin.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/auth/register</b>            | <b>auth.js</b>  | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/auth/verify-email</b>        | <b>auth.js</b>  | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/auth/login</b>               | <b>auth.js</b>  | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/auth/resend-verification</b> | <b>auth.js</b>  | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/auth/logout</b>              | <b>auth.js</b>  | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/auth/forgot-password</b>     | <b>auth.js</b>  | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/auth/reset-password</b>      | <b>auth.js</b>  | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/auth/me</b>                  | <b>auth.js</b>  | <b>Secured (JWT Required)</b> |

|              |                                     |                      |                               |
|--------------|-------------------------------------|----------------------|-------------------------------|
| <b>GET</b>   | <b>/api/auth/admin-test</b>         | <b>auth.js</b>       | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/chatbot/ask</b>             | <b>chatbot.js</b>    | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/complaints/</b>             | <b>complaints.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/complaints/:id/messages</b> | <b>complaints.js</b> | <b>Secured (JWT Required)</b> |
| <b>PATCH</b> | <b>/api/complaints/:id/status</b>   | <b>complaints.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>   | <b>/api/complaints/</b>             | <b>complaints.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>   | <b>/api/complaints/:id</b>          | <b>complaints.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>   | <b>/api/config/shop-status</b>      | <b>config.js</b>     | <b>Public Access</b>          |
| <b>GET</b>   | <b>/api/config/limits</b>           | <b>config.js</b>     | <b>Public Access</b>          |
| <b>GET</b>   | <b>/api/config/push</b>             | <b>config.js</b>     | <b>Public Access</b>          |
| <b>POST</b>  | <b>/api/contact/</b>                | <b>contact.js</b>    | <b>Secured (JWT Required)</b> |
| <b>GET</b>   | <b>/api/contact/</b>                | <b>contact.js</b>    | <b>Secured (JWT Required)</b> |

|               |                                       |                            |                               |
|---------------|---------------------------------------|----------------------------|-------------------------------|
| <b>PATCH</b>  | <b>/api/contact/:id/status</b>        | <b>contact.js</b>          | <b>Secured (JWT Required)</b> |
| <b>PATCH</b>  | <b>/api/contact/:id/respond</b>       | <b>contact.js</b>          | <b>Secured (JWT Required)</b> |
| <b>DELETE</b> | <b>/api/contact/:id</b>               | <b>contact.js</b>          | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/convertRoutes/docx</b>        | <b>convertRoutes.js</b>    | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/documents/upload</b>          | <b>documents.js</b>        | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/documents/upload-multiple</b> | <b>documents.js</b>        | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/documents/library</b>         | <b>documents.js</b>        | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/documents/:id/url</b>         | <b>documents.js</b>        | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/friendshipRoutes/search</b>   | <b>friendshipRoutes.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/friendshipRoutes/request</b>  | <b>friendshipRoutes.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/friendshipRoutes/respond</b>  | <b>friendshipRoutes.js</b> | <b>Secured (JWT Required)</b> |

|               |                                                         |                                 |                                   |
|---------------|---------------------------------------------------------|---------------------------------|-----------------------------------|
| <b>GET</b>    | <b>/api/friendshipRoutes/</b>                           | <b>friendshipRoute<br/>s.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>    | <b>/api/friendshipRoutes/pending</b>                    | <b>friendshipRoute<br/>s.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>DELETE</b> | <b>/api/friendshipRoutes/:friendsh<br/>ipId</b>         | <b>friendshipRoute<br/>s.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>   | <b>/api/groupChatRoutes/</b>                            | <b>groupChatRoute<br/>s.js</b>  | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>    | <b>/api/groupChatRoutes/</b>                            | <b>groupChatRoute<br/>s.js</b>  | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>    | <b>/api/groupChatRoutes/:groupId<br/>/messages</b>      | <b>groupChatRoute<br/>s.js</b>  | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>   | <b>/api/groupChatRoutes/:groupId<br/>/messages</b>      | <b>groupChatRoute<br/>s.js</b>  | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>   | <b>/api/groupChatRoutes/:groupId<br/>/send-voice</b>    | <b>groupChatRoute<br/>s.js</b>  | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>   | <b>/api/groupChatRoutes/:groupId<br/>/add-member</b>    | <b>groupChatRoute<br/>s.js</b>  | <b>Secured (JWT<br/>Required)</b> |
| <b>DELETE</b> | <b>/api/groupChatRoutes/:groupId<br/>/remove-member</b> | <b>groupChatRoute<br/>s.js</b>  | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>   | <b>/api/groupChatRoutes/:groupId<br/>/leave</b>         | <b>groupChatRoute<br/>s.js</b>  | <b>Secured (JWT<br/>Required)</b> |

|             |                                                       |                                  |                                   |
|-------------|-------------------------------------------------------|----------------------------------|-----------------------------------|
| <b>POST</b> | <b>/api/groupChatRoutes/:groupId<br/>/disband</b>     | <b>groupChatRoute<br/>s.js</b>   | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b> | <b>/api/groupChatRoutes/:groupId<br/>/delete-chat</b> | <b>groupChatRoute<br/>s.js</b>   | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>  | <b>/api/groupOrderRoutes/</b>                         | <b>groupOrderRoute<br/>es.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>  | <b>/api/groupOrderRoutes/my-<br/>active</b>           | <b>groupOrderRoute<br/>es.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>  | <b>/api/groupOrderRoutes/group-<br/>chat/:groupId</b> | <b>groupOrderRoute<br/>es.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b> | <b>/api/groupOrderRoutes/custom-<br/>split</b>        | <b>groupOrderRoute<br/>es.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b> | <b>/api/groupOrderRoutes/split/cu<br/>stom</b>        | <b>groupOrderRoute<br/>es.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b> | <b>/api/groupOrderRoutes/split/se<br/>nd</b>          | <b>groupOrderRoute<br/>es.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b> | <b>/api/groupOrderRoutes/split/re<br/>spond</b>       | <b>groupOrderRoute<br/>es.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>  | <b>/api/groupOrderRoutes/split/pe<br/>nding</b>       | <b>groupOrderRoute<br/>es.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>  | <b>/api/groupOrderRoutes/:orderId</b>                 | <b>groupOrderRoute<br/>es.js</b> | <b>Secured (JWT<br/>Required)</b> |

|              |                                                              |                                 |                                   |
|--------------|--------------------------------------------------------------|---------------------------------|-----------------------------------|
| <b>GET</b>   | <b>/api/groupOrderRoutes/:group<br/>OrderId/split-status</b> | <b>groupOrderRout<br/>es.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>  | <b>/api/guest/request-otp</b>                                | <b>guest.js</b>                 | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>  | <b>/api/guest/verify-otp</b>                                 | <b>guest.js</b>                 | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>  | <b>/api/guest/end-session</b>                                | <b>guest.js</b>                 | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>   | <b>/api/inventory/</b>                                       | <b>inventory.js</b>             | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>   | <b>/api/inventory/summary</b>                                | <b>inventory.js</b>             | <b>Secured (JWT<br/>Required)</b> |
| <b>PATCH</b> | <b>/api/inventory/:id/update</b>                             | <b>inventory.js</b>             | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>  | <b>/api/messageRoutes/send</b>                               | <b>messageRoutes.j<br/>s</b>    | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>  | <b>/api/messageRoutes/send-<br/>voice/:receiverId</b>        | <b>messageRoutes.j<br/>s</b>    | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>   | <b>/api/messageRoutes/unread-<br/>count</b>                  | <b>messageRoutes.j<br/>s</b>    | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>   | <b>/api/messageRoutes/conversatio<br/>ns</b>                 | <b>messageRoutes.j<br/>s</b>    | <b>Secured (JWT<br/>Required)</b> |

|               |                                          |                         |                               |
|---------------|------------------------------------------|-------------------------|-------------------------------|
| <b>GET</b>    | <b>/api/messageRoutes/:friendId</b>      | <b>messageRoutes.js</b> | <b>Secured (JWT Required)</b> |
| <b>PUT</b>    | <b>/api/messageRoutes/:friendId/read</b> | <b>messageRoutes.js</b> | <b>Secured (JWT Required)</b> |
| <b>DELETE</b> | <b>/api/messageRoutes/:messageId</b>     | <b>messageRoutes.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/notifications/</b>               | <b>notifications.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/notifications/all</b>            | <b>notifications.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/notifications/unread-count</b>   | <b>notifications.js</b> | <b>Secured (JWT Required)</b> |
| <b>PATCH</b>  | <b>/api/notifications/:id/read</b>       | <b>notifications.js</b> | <b>Secured (JWT Required)</b> |
| <b>PATCH</b>  | <b>/api/notifications/read-all</b>       | <b>notifications.js</b> | <b>Secured (JWT Required)</b> |
| <b>DELETE</b> | <b>/api/notifications/:id</b>            | <b>notifications.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/orders/estimate</b>              | <b>orders.js</b>        | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/orders/track/:orderId</b>        | <b>orders.js</b>        | <b>Secured (JWT Required)</b> |

|              |                                   |                    |                               |
|--------------|-----------------------------------|--------------------|-------------------------------|
| <b>GET</b>   | <b>/api/orders/my-orders</b>      | <b>orders.js</b>   | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/orders/create</b>         | <b>orders.js</b>   | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/orders/:id/reorder</b>    | <b>orders.js</b>   | <b>Secured (JWT Required)</b> |
| <b>GET</b>   | <b>/api/orders/:id</b>            | <b>orders.js</b>   | <b>Secured (JWT Required)</b> |
| <b>GET</b>   | <b>/api/orders/:id/wait</b>       | <b>orders.js</b>   | <b>Secured (JWT Required)</b> |
| <b>GET</b>   | <b>/api/orders/:id/receipt</b>    | <b>orders.js</b>   | <b>Secured (JWT Required)</b> |
| <b>PATCH</b> | <b>/api/orders/:id/cancel</b>     | <b>orders.js</b>   | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/payments/webhook</b>      | <b>payments.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/payments/create-order</b> | <b>payments.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/payments/verify</b>       | <b>payments.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/payments/cash-confirm</b> | <b>payments.js</b> | <b>Secured (JWT Required)</b> |

|             |                                       |                         |                               |
|-------------|---------------------------------------|-------------------------|-------------------------------|
| <b>POST</b> | <b>/api/ratings/</b>                  | <b>ratings.js</b>       | <b>Secured (JWT Required)</b> |
| <b>GET</b>  | <b>/api/ratings/my-ratings</b>        | <b>ratings.js</b>       | <b>Secured (JWT Required)</b> |
| <b>GET</b>  | <b>/api/receiptRoutes/:orderId</b>    | <b>receiptRoutes.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>  | <b>/api/reports/daily-orders</b>      | <b>reports.js</b>       | <b>Secured (JWT Required)</b> |
| <b>GET</b>  | <b>/api/reports/revenue</b>           | <b>reports.js</b>       | <b>Secured (JWT Required)</b> |
| <b>GET</b>  | <b>/api/reports/service-breakdown</b> | <b>reports.js</b>       | <b>Secured (JWT Required)</b> |
| <b>GET</b>  | <b>/api/reports/peak-hours</b>        | <b>reports.js</b>       | <b>Secured (JWT Required)</b> |
| <b>GET</b>  | <b>/api/reports/staff-performance</b> | <b>reports.js</b>       | <b>Secured (JWT Required)</b> |
| <b>GET</b>  | <b>/api/reports/payment-breakdown</b> | <b>reports.js</b>       | <b>Secured (JWT Required)</b> |
| <b>GET</b>  | <b>/api/reports/export</b>            | <b>reports.js</b>       | <b>Secured (JWT Required)</b> |
| <b>GET</b>  | <b>/api/staff/queue</b>               | <b>staff.js</b>         | <b>Secured (JWT Required)</b> |

|              |                                                    |                        |                                   |
|--------------|----------------------------------------------------|------------------------|-----------------------------------|
| <b>GET</b>   | <b>/api/staff/staff/orders/complete<br/>d</b>      | <b>staff.js</b>        | <b>Secured (JWT<br/>Required)</b> |
| <b>PATCH</b> | <b>/api/staff/orders/:id/lock</b>                  | <b>staff.js</b>        | <b>Secured (JWT<br/>Required)</b> |
| <b>PATCH</b> | <b>/api/staff/orders/:id/unlock</b>                | <b>staff.js</b>        | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>  | <b>/api/staff/orders/:id/start-<br/>processing</b> | <b>staff.js</b>        | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>  | <b>/api/staff/orders/:id/ready-for-<br/>pickup</b> | <b>staff.js</b>        | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>  | <b>/api/staff/orders/:id/collect-<br/>cash</b>     | <b>staff.js</b>        | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>  | <b>/api/staff/orders/:id/verify-otp</b>            | <b>staff.js</b>        | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>  | <b>/api/staff/orders/:id/partial</b>               | <b>staff.js</b>        | <b>Secured (JWT<br/>Required)</b> |
| <b>POST</b>  | <b>/api/staffReports/</b>                          | <b>staffReports.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>GET</b>   | <b>/api/staffReports/</b>                          | <b>staffReports.js</b> | <b>Secured (JWT<br/>Required)</b> |
| <b>PATCH</b> | <b>/api/staffReports/:id/acknowled<br/>ge</b>      | <b>staffReports.js</b> | <b>Secured (JWT<br/>Required)</b> |

|              |                                         |                        |                               |
|--------------|-----------------------------------------|------------------------|-------------------------------|
| <b>PATCH</b> | <b>/api/staffReports/:id/resolve</b>    | <b>staffReports.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/tools/convert</b>               | <b>tools.js</b>        | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/tools/resize-image</b>          | <b>tools.js</b>        | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/tools/compress-image</b>        | <b>tools.js</b>        | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/tools/resize-pdf-pages</b>      | <b>tools.js</b>        | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/uploadRoutes/</b>               | <b>uploadRoutes.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>   | <b>/api/users/me</b>                    | <b>users.js</b>        | <b>Secured (JWT Required)</b> |
| <b>PATCH</b> | <b>/api/users/me</b>                    | <b>users.js</b>        | <b>Secured (JWT Required)</b> |
| <b>PATCH</b> | <b>/api/users/me/change-password</b>    | <b>users.js</b>        | <b>Secured (JWT Required)</b> |
| <b>POST</b>  | <b>/api/users/me/logout-all-devices</b> | <b>users.js</b>        | <b>Secured (JWT Required)</b> |
| <b>GET</b>   | <b>/api/users/me/usage-summary</b>      | <b>users.js</b>        | <b>Secured (JWT Required)</b> |

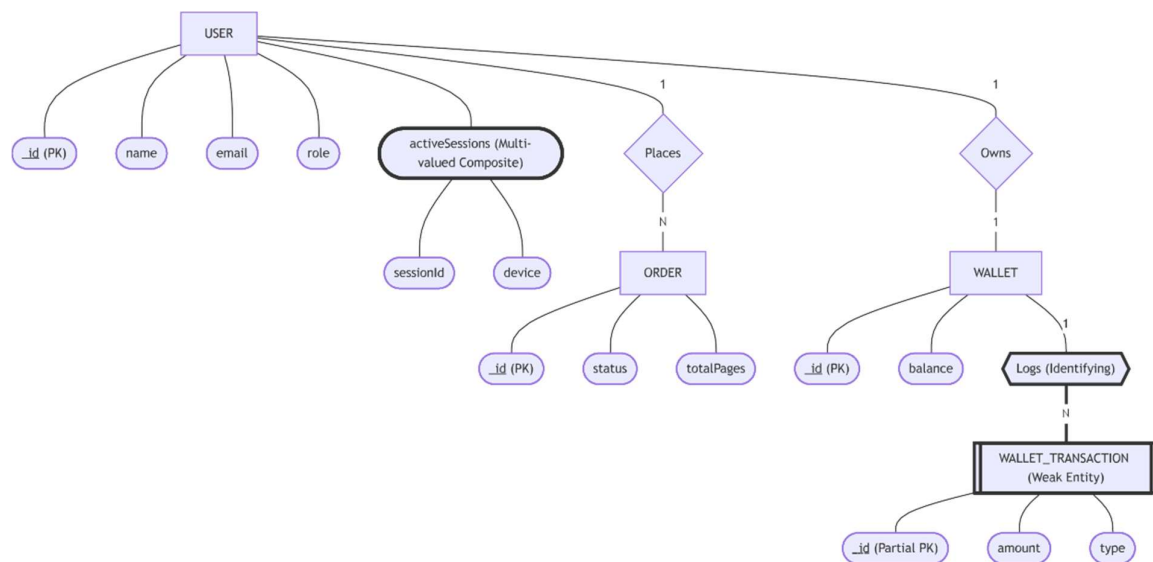
|               |                                              |                        |                               |
|---------------|----------------------------------------------|------------------------|-------------------------------|
| <b>POST</b>   | <b>/api/users/push-subscription</b>          | <b>users.js</b>        | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/users/test-push</b>                  | <b>users.js</b>        | <b>Secured (JWT Required)</b> |
| <b>DELETE</b> | <b>/api/users/push-subscription</b>          | <b>users.js</b>        | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/walletRoutes/</b>                    | <b>walletRoutes.js</b> | <b>Secured (JWT Required)</b> |
| <b>GET</b>    | <b>/api/walletRoutes/transactions</b>        | <b>walletRoutes.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/walletRoutes/topup/create-order</b>  | <b>walletRoutes.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/walletRoutes/topup/verify</b>        | <b>walletRoutes.js</b> | <b>Secured (JWT Required)</b> |
| <b>POST</b>   | <b>/api/walletRoutes/complete-onboarding</b> | <b>walletRoutes.js</b> | <b>Secured (JWT Required)</b> |

### 3. Database Design & Data Models

The system is designed to utilize a NoSQL document-based architecture via MongoDB. Data integrity will be maintained securely using Mongoose schemas. Below is the proposed structural design of the core database entities.

#### Complete Entity-Relationship (ER) Conceptual Model (Chen's Notation)

This ER model strictly adheres to conceptual design rules: independent entities (rectangles), attributes (ovals), active relationships (diamonds), and correctly mapped structural constraints. It accurately reflects the actual MongoDB schema without injecting fake constraints.



### 3.ActivityLog Planned Schema: **ActivityLog**

| Proposed Field Name | Data Type / Constraints | Expected Behavior            |
|---------------------|-------------------------|------------------------------|
| actionType          | String                  | Required                     |
| performedBy         | ObjectId (Foreign Key)  | Optional                     |
| description         | String                  | Required                     |
| affectedRecordId    | ObjectId (Foreign Key)  | Optional                     |
| metadata            | Mixed Object            | Optional                     |
| timestamp           | Date                    | Optional   Default: Date.now |
| type                | String                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | Date                    | Base Property Configuration  |
| default             | Date                    | Base Property Configuration  |

### 3.Complaint Planned Schema: **Complaint**

| Proposed Field Name | Data Type / Constraints | Expected Behavior            |
|---------------------|-------------------------|------------------------------|
| status              | String                  | Required                     |
| timestamp           | Date                    | Optional   Default: Date.now |
| actorId             | ObjectId (Foreign Key)  | Optional   Default: null     |
| actorRole           | String                  | Optional   Default: null     |
| note                | String                  | Optional   Default: "        |
| senderId            | ObjectId (Foreign Key)  | Optional   Default: null     |
| senderRole          | String                  | Optional   Default: null     |
| text                | String                  | Required                     |
| attachmentUrl       | String                  | Optional   Default: "        |
| timestamp           | Date                    | Optional   Default: Date.now |
| complaintToken      | String                  | Required                     |
| orderId             | ObjectId (Foreign Key)  | Required                     |
| raisedBy            | ObjectId (Foreign Key)  | Required                     |
| category            | String                  | Required                     |
| description         | String                  | Required                     |
| status              | String                  | Optional   Default: 'Open'   |
| statusHistory       | Mixed Object            | Optional   Default: []       |
| messages            | Mixed Object            | Optional   Default: []       |
| attachmentUrls      | Mixed Object            | Optional   Default: []       |

|            |        |                             |
|------------|--------|-----------------------------|
| resolvedAt | Date   | Optional   Default: null    |
| type       | String | Base Property Configuration |
| type       | Date   | Base Property Configuration |
| default    | Date   | Base Property Configuration |
| type       | String | Base Property Configuration |
| type       | String | Base Property Configuration |
| type       | String | Base Property Configuration |
| type       | String | Base Property Configuration |
| type       | String | Base Property Configuration |
| type       | Date   | Base Property Configuration |
| default    | Date   | Base Property Configuration |
| type       | String | Base Property Configuration |
| type       | String | Base Property Configuration |
| type       | String | Base Property Configuration |
| type       | String | Base Property Configuration |
| type       | Date   | Base Property Configuration |

### 3.DailyCounter Planned Schema: `DailyCounter`

| Proposed Field Name | Data Type / Constraints | Expected Behavior           |
|---------------------|-------------------------|-----------------------------|
| date                | String                  | Required                    |
| prefix              | String                  | Required   Default: 'Order' |
| count               | Number                  | Optional   Default: 0       |
| type                | String                  | Base Property Configuration |
| type                | String                  | Base Property Configuration |
| type                | Number                  | Base Property Configuration |

### 3.Document Planned Schema: Document

| Proposed Field Name | Data Type / Constraints | Expected Behavior            |
|---------------------|-------------------------|------------------------------|
| page                | Number                  | Optional                     |
| issue               | String                  | Optional                     |
| blankPages          | Mixed Object            | Optional   Default: []       |
| qualityIssues       | Mixed Object            | Optional   Default: []       |
| isPasswordProtected | Boolean                 | Optional   Default: false    |
| colourHeavyPages    | Mixed Object            | Optional   Default: []       |
| cloudinaryUrl       | String                  | Required                     |
| publicId            | String                  | Required                     |
| originalFilename    | String                  | Required                     |
| fileType            | String                  | Required                     |
| pageCount           | Number                  | Optional                     |
| analysisResults     | Mixed Object            | Optional   Default: () => ({ |
| ownerId             | ObjectId (Foreign Key)  | Optional                     |
| isGuest             | Boolean                 | Optional   Default: false    |
| guestSessionId      | String                  | Optional                     |
| type                | Number                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | Boolean                 | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |

### 3.Friendship Planned Schema: **Friendship**

| Proposed Field Name | Data Type / Constraints | Expected Behavior             |
|---------------------|-------------------------|-------------------------------|
| requester           | ObjectId (Foreign Key)  | Required                      |
| recipient           | ObjectId (Foreign Key)  | Required                      |
| status              | String                  | Optional   Default: 'pending' |
| createdAt           | Date                    | Optional   Default: Date.now  |
| updatedAt           | Date                    | Optional   Default: Date.now  |
| type                | String                  | Base Property Configuration   |
| type                | Date                    | Base Property Configuration   |
| default             | Date                    | Base Property Configuration   |
| type                | Date                    | Base Property Configuration   |
| default             | Date                    | Base Property Configuration   |

### 3.Group Planned Schema: **Group**

| Proposed Field Name | Data Type / Constraints | Expected Behavior            |
|---------------------|-------------------------|------------------------------|
| name                | String                  | Required                     |
| createdBy           | ObjectId (Foreign Key)  | Required                     |
| members             | ObjectId (Foreign Key)  | Optional                     |
| validate            | Mixed Object            | Optional                     |
| isDisbanded         | Boolean                 | Optional   Default: false    |
| createdAt           | Date                    | Optional   Default: Date.now |
| type                | String                  | Base Property Configuration  |
| type                | Boolean                 | Base Property Configuration  |
| type                | Date                    | Base Property Configuration  |
| default             | Date                    | Base Property Configuration  |

### 3.GroupOrder Planned Schema: GroupOrder

| Proposed Field Name     | Data Type / Constraints | Expected Behavior             |
|-------------------------|-------------------------|-------------------------------|
| userId                  | ObjectId (Foreign Key)  | Required                      |
| amount                  | Number                  | Required                      |
| walletStatus            | String                  | Optional   Default: 'pending' |
| paidAt                  | Date                    | Optional   Default: null      |
| orderId                 | ObjectId (Foreign Key)  | Optional   Default: null      |
| createdBy               | ObjectId (Foreign Key)  | Required                      |
| totalAmount             | Number                  | Required                      |
| participants            | Mixed Object            | Required                      |
| description             | String                  | Optional   Default: "         |
| status                  | String                  | Optional   Default: 'active'  |
| createdAt               | Date                    | Optional   Default: Date.now  |
| partialFilterExpression | Mixed Object            | Optional                      |
| type                    | Number                  | Base Property Configuration   |
| type                    | String                  | Base Property Configuration   |
| type                    | Date                    | Base Property Configuration   |
| type                    | Number                  | Base Property Configuration   |
| type                    | String                  | Base Property Configuration   |
| type                    | String                  | Base Property Configuration   |
| type                    | Date                    | Base Property Configuration   |
| default                 | Date                    | Base Property Configuration   |

### 3. GuestSession Planned Schema: GuestSession

| Proposed Field Name   | Data Type / Constraints | Expected Behavior            |
|-----------------------|-------------------------|------------------------------|
| email                 | String                  | Required                     |
| hashedOtp             | String                  | Optional                     |
| otpExpiry             | Date                    | Optional                     |
| otpRequestCount       | Number                  | Optional   Default: 0        |
| otpRequestWindowStart | Date                    | Optional                     |
| sessionId             | String                  | Optional   Default: uuid     |
| sessionToken          | String                  | Optional                     |
| isActive              | Boolean                 | Optional   Default: true     |
| createdAt             | Date                    | Optional   Default: Date.now |
| type                  | String                  | Base Property Configuration  |
| type                  | String                  | Base Property Configuration  |
| type                  | Date                    | Base Property Configuration  |
| type                  | Number                  | Base Property Configuration  |
| type                  | Date                    | Base Property Configuration  |
| type                  | String                  | Base Property Configuration  |
| type                  | String                  | Base Property Configuration  |
| type                  | Boolean                 | Base Property Configuration  |
| type                  | Date                    | Base Property Configuration  |
| default               | Date                    | Base Property Configuration  |

### 3. Inquiry Planned Schema: Inquiry

| Proposed Field Name | Data Type / Constraints | Expected Behavior             |
|---------------------|-------------------------|-------------------------------|
| name                | String                  | Optional                      |
| email               | String                  | Optional                      |
| subject             | String                  | Optional                      |
| message             | String                  | Optional                      |
| status              | String                  | Optional   Default: 'Pending' |
| response            | String                  | Optional                      |
| respondedAt         | Date                    | Optional                      |
| respondedBy         | ObjectId (Foreign Key)  | Optional                      |
| type                | String                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |
| type                | Date                    | Base Property Configuration   |

### 3.InventoryItem Planned Schema: `InventoryItem`

| Proposed Field Name               | Data Type / Constraints | Expected Behavior           |
|-----------------------------------|-------------------------|-----------------------------|
| <code>name</code>                 | String                  | Required                    |
| <code>category</code>             | String                  | Required                    |
| <code>unit</code>                 | String                  | Required                    |
| <code>currentStock</code>         | Number                  | Required                    |
| <code>minimumThreshold</code>     | Number                  | Required                    |
| <code>usageRate</code>            | Number                  | Optional   Default: 0       |
| <code>daysOfStockRemaining</code> | Number                  | Optional   Default: 0       |
| <code>lastUpdatedBy</code>        | ObjectId (Foreign Key)  | Optional   Default: null    |
| <code>lastUpdatedAt</code>        | Date                    | Optional   Default: null    |
| <code>type</code>                 | String                  | Base Property Configuration |
| <code>type</code>                 | String                  | Base Property Configuration |
| <code>type</code>                 | String                  | Base Property Configuration |
| <code>type</code>                 | Number                  | Base Property Configuration |
| <code>type</code>                 | Number                  | Base Property Configuration |
| <code>type</code>                 | Number                  | Base Property Configuration |
| <code>type</code>                 | Number                  | Base Property Configuration |
| <code>type</code>                 | Date                    | Base Property Configuration |

### 3.Message Planned Schema: **Message**

| Proposed Field Name | Data Type / Constraints | Expected Behavior            |
|---------------------|-------------------------|------------------------------|
| senderId            | ObjectId (Foreign Key)  | Required                     |
| recipientId         | ObjectId (Foreign Key)  | Optional   Default: null     |
| groupId             | ObjectId (Foreign Key)  | Optional   Default: null     |
| content             | String                  | Optional   Default: ""       |
| mediaUrl            | String                  | Optional   Default: null     |
| mediaType           | String                  | Optional   Default: 'none'   |
| read                | Boolean                 | Optional   Default: false    |
| type                | String                  | Optional   Default: 'text'   |
| audioUrl            | String                  | Optional   Default: null     |
| audioDuration       | Number                  | Optional   Default: null     |
| splitCardData       | Mixed Object            | Optional   Default: null     |
| isSystem            | Boolean                 | Optional   Default: false    |
| isDeleted           | Boolean                 | Optional   Default: false    |
| timestamp           | Date                    | Optional   Default: Date.now |
| type                | String                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | Boolean                 | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | Number                  | Base Property Configuration  |
| type                | Boolean                 | Base Property Configuration  |
| type                | Boolean                 | Base Property Configuration  |
| type                | Date                    | Base Property Configuration  |
| default             | Date                    | Base Property Configuration  |

### 3.Notification Planned Schema: Notification

| Proposed Field Name | Data Type / Constraints | Expected Behavior            |
|---------------------|-------------------------|------------------------------|
| recipientId         | ObjectId (Foreign Key)  | Optional   Default: null     |
| recipientRole       | String                  | Optional   Default: null     |
| title               | String                  | Required                     |
| message             | String                  | Required                     |
| type                | String                  | Required   Default: 'system' |
| recipientType       | String                  | Required                     |
| urgency             | String                  | Optional   Default: 'Normal' |
| relatedEntity       | String                  | Optional                     |
| isRead              | Boolean                 | Optional   Default: false    |
| relatedOrderId      | ObjectId (Foreign Key)  | Optional   Default: null     |
| relatedServiceType  | String                  | Optional   Default: null     |
| type                | String                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | Boolean                 | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |

### 3.Order Planned Schema: Order

| Proposed Field Name | Data Type / Constraints | Expected Behavior                   |
|---------------------|-------------------------|-------------------------------------|
| status              | String                  | Optional                            |
| actorId             | ObjectId (Foreign Key)  | Optional                            |
| actorRole           | String                  | Optional                            |
| timestamp           | Date                    | Optional   Default: Date.now        |
| note                | String                  | Optional                            |
| copies              | Number                  | Optional   Default: 1               |
| colourMode          | String                  | Optional   Default: 'BlackAndWhite' |
| sided               | String                  | Optional   Default: 'Single'        |
| paperSize           | String                  | Optional   Default: 'A4'            |
| binding             | String                  | Optional   Default: 'None'          |
| printInstructions   | String                  | Optional                            |
| outputFormat        | String                  | Optional                            |
| conversionType      | String                  | Optional                            |
| userId              | ObjectId (Foreign Key)  | Optional                            |
| isGuest             | Boolean                 | Optional   Default: false           |
| guestEmail          | String                  | Optional                            |
| guestSessionId      | String                  | Optional                            |
| serviceType         | String                  | Required                            |
| userRole            | String                  | Required                            |
| printConfig         | Mixed Object            | Optional   Default: {} => ({        |

|                       |                        |                               |
|-----------------------|------------------------|-------------------------------|
| pageCount             | Number                 | Optional                      |
| estimatedCost         | Number                 | Optional                      |
| finalCost             | Number                 | Optional                      |
| paymentMethod         | String                 | Optional                      |
| paymentStatus         | String                 | Optional   Default: 'Pending' |
| razorpayOrderId       | String                 | Optional                      |
| razorpayPaymentId     | String                 | Optional                      |
| status                | String                 | Optional   Default: 'Pending' |
| statusHistory         | Mixed Object           | Optional   Default: []        |
| tokenNumber           | String                 | Optional                      |
| priorityScore         | Number                 | Optional   Default: 0         |
| agingScore            | Number                 | Optional   Default: 0         |
| preferredPickupSlot   | String                 | Optional                      |
| assignedStaffId       | ObjectId (Foreign Key) | Optional                      |
| pickupOtp             | String                 | Optional                      |
| pickupOtpExpiry       | Date                   | Optional                      |
| pickupOtpVerified     | Boolean                | Optional   Default: false     |
| otpAttempts           | Number                 | Optional   Default: 0         |
| otpLocked             | Boolean                | Optional   Default: false     |
| lockedBy              | ObjectId (Foreign Key) | Optional                      |
| lockedAt              | Date                   | Optional                      |
| partialPagesCompleted | Number                 | Optional   Default: 0         |

|                                |                        |                             |
|--------------------------------|------------------------|-----------------------------|
| <code>followUpOrderId</code>   | ObjectId (Foreign Key) | Optional                    |
| <code>internalNotes</code>     | String                 | Optional   Default: "       |
| <code>estimatedDuration</code> | Number                 | Optional                    |
| <code>cancelReason</code>      | String                 | Optional                    |
| <code>isGroupOrder</code>      | Boolean                | Optional   Default: false   |
| <code>groupOrderId</code>      | ObjectId (Foreign Key) | Optional   Default: null    |
| <code>type</code>              | String                 | Base Property Configuration |
| <code>type</code>              | String                 | Base Property Configuration |
| <code>type</code>              | Date                   | Base Property Configuration |
| <code>default</code>           | Date                   | Base Property Configuration |
| <code>type</code>              | String                 | Base Property Configuration |
| <code>type</code>              | Number                 | Base Property Configuration |
| <code>type</code>              | String                 | Base Property Configuration |
| <code>type</code>              | String                 | Base Property Configuration |
| <code>type</code>              | String                 | Base Property Configuration |
| <code>type</code>              | String                 | Base Property Configuration |
| <code>type</code>              | String                 | Base Property Configuration |
| <code>type</code>              | String                 | Base Property Configuration |
| <code>type</code>              | String                 | Base Property Configuration |
| <code>type</code>              | Boolean                | Base Property Configuration |
| <code>type</code>              | String                 | Base Property Configuration |

|      |         |                             |
|------|---------|-----------------------------|
| type | String  | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | String  | Base Property Configuration |
| type | String  | Base Property Configuration |
| type | Date    | Base Property Configuration |
| type | Boolean | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | Boolean | Base Property Configuration |
| type | Date    | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | String  | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | String  | Base Property Configuration |
| type | Boolean | Base Property Configuration |

### 3.Payment Planned Schema: **Payment**

| Proposed Field Name | Data Type / Constraints | Expected Behavior             |
|---------------------|-------------------------|-------------------------------|
| orderId             | ObjectId (Foreign Key)  | Required                      |
| userId              | ObjectId (Foreign Key)  | Optional                      |
| amount              | Number                  | Required                      |
| currency            | String                  | Optional   Default: 'INR'     |
| razorpayOrderId     | String                  | Optional                      |
| razorpayPaymentId   | String                  | Optional                      |
| razorpaySignature   | String                  | Optional                      |
| status              | String                  | Optional   Default: 'Created' |
| method              | String                  | Optional                      |
| type                | Number                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |
| type                | String                  | Base Property Configuration   |

### 3.Rating Planned Schema: Rating

| Proposed Field Name | Data Type / Constraints | Expected Behavior            |
|---------------------|-------------------------|------------------------------|
| orderId             | ObjectId (Foreign Key)  | Required                     |
| userId              | ObjectId (Foreign Key)  | Required                     |
| serviceType         | String                  | Required                     |
| stars               | Number                  | Required                     |
| comment             | String                  | Optional                     |
| createdAt           | Date                    | Optional   Default: Date.now |
| type                | String                  | Base Property Configuration  |
| type                | Number                  | Base Property Configuration  |
| type                | String                  | Base Property Configuration  |
| type                | Date                    | Base Property Configuration  |
| default             | Date                    | Base Property Configuration  |

### 3.SplitRequest Planned Schema: `SplitRequest`

| Proposed Field Name        | Data Type / Constraints | Expected Behavior             |
|----------------------------|-------------------------|-------------------------------|
| <code>groupOrderId</code>  | ObjectId (Foreign Key)  | Required                      |
| <code>from</code>          | ObjectId (Foreign Key)  | Required                      |
| <code>to</code>            | ObjectId (Foreign Key)  | Required                      |
| <code>amount</code>        | Number                  | Required                      |
| <code>status</code>        | String                  | Optional   Default: 'pending' |
| <code>triggeredFrom</code> | String                  | Required                      |
| <code>groupChatId</code>   | ObjectId (Foreign Key)  | Optional   Default: null      |
| <code>createdAt</code>     | Date                    | Optional   Default: Date.now  |
| <code>resolvedAt</code>    | Date                    | Optional   Default: null      |
| <code>type</code>          | Number                  | Base Property Configuration   |
| <code>type</code>          | String                  | Base Property Configuration   |
| <code>type</code>          | String                  | Base Property Configuration   |
| <code>type</code>          | Date                    | Base Property Configuration   |
| <code>default</code>       | Date                    | Base Property Configuration   |
| <code>type</code>          | Date                    | Base Property Configuration   |

### 3.StaffReport Planned Schema: StaffReport

| Proposed Field Name   | Data Type / Constraints | Expected Behavior             |
|-----------------------|-------------------------|-------------------------------|
| reportRef             | String                  | Required                      |
| raisedBy              | ObjectId (Foreign Key)  | Required                      |
| category              | String                  | Required                      |
| specificItem          | String                  | Optional                      |
| description           | String                  | Required                      |
| urgency               | String                  | Optional   Default: 'Normal'  |
| status                | String                  | Optional   Default: 'Pending' |
| resolutionNote        | String                  | Optional                      |
| acknowledgedAt        | Date                    | Optional                      |
| resolvedAt            | Date                    | Optional                      |
| linkedInventoryItemId | ObjectId (Foreign Key)  | Optional                      |
| type                  | String                  | Base Property Configuration   |
| type                  | String                  | Base Property Configuration   |
| type                  | String                  | Base Property Configuration   |
| type                  | String                  | Base Property Configuration   |
| type                  | String                  | Base Property Configuration   |
| type                  | String                  | Base Property Configuration   |
| type                  | String                  | Base Property Configuration   |
| type                  | Date                    | Base Property Configuration   |
| type                  | Date                    | Base Property Configuration   |

### 3.SystemConfig Planned Schema: SystemConfig

| Proposed Field Name   | Data Type / Constraints | Expected Behavior                                         |
|-----------------------|-------------------------|-----------------------------------------------------------|
| Monday                | Mixed Object            | Optional                                                  |
| Tuesday               | Mixed Object            | Optional                                                  |
| Wednesday             | Mixed Object            | Optional                                                  |
| Thursday              | Mixed Object            | Optional                                                  |
| Friday                | Mixed Object            | Optional                                                  |
| open                  | String                  | Required                                                  |
| close                 | String                  | Required                                                  |
| pricing               | Number                  | Optional   Default: DEFAULT_PRICING.printBW               |
| printColour           | Number                  | Optional   Default: DEFAULT_PRICING.printColour           |
| doubleSidedMultiplier | Number                  | Optional   Default: DEFAULT_PRICING.doubleSidedMultiplier |
| photocopy             | Number                  | Optional   Default: DEFAULT_PRICING.photocopy             |
| scanning              | Number                  | Optional   Default: DEFAULT_PRICING.scanning              |
| bindingSpiral         | Number                  | Optional   Default: DEFAULT_PRICING.bindingSpiral         |
| bindingStaple         | Number                  | Optional   Default: DEFAULT_PRICING.bindingStaple         |
| conversion            | Number                  | Optional   Default: DEFAULT_PRICING.conversion            |
| printTimePerPage      | Number                  | Optional   Default: DEFAULT_PRICING.printTimePerPage      |
| photocopyTimePerPage  | Number                  | Optional   Default: DEFAULT_PRICING.photocopyTimePerPage  |
| scanTimePerPage       | Number                  | Optional   Default: DEFAULT_PRICING.scanTimePerPage       |
| bindingSpiralDuration | Number                  | Optional   Default: DEFAULT_PRICING.bindingSpiralDuration |

|                                         |              |                                                                            |
|-----------------------------------------|--------------|----------------------------------------------------------------------------|
| <code>bindingStapleDuration</code>      | Number       | Optional   Default: DEFAULT_PRICING.bindingStapleDuration                  |
| <code>conversionDuration</code>         | Number       | Optional   Default: DEFAULT_PRICING.conversionDuration                     |
| <code>orderLimits</code>                | Number       | Optional   Default:<br>DEFAULT_ORDER_LIMITS.maxPagesPerOrder               |
| <code>maxCopies</code>                  | Number       | Optional   Default: DEFAULT_ORDER_LIMITS.maxCopies                         |
| <code>maxDocumentsPerOrder</code>       | Number       | Optional   Default:<br>DEFAULT_ORDER_LIMITS.maxDocumentsPerOrder           |
| <code>maxOrdersPerUserPerDay</code>     | Number       | Optional   Default:<br>DEFAULT_ORDER_LIMITS.maxOrdersPerUserPerDay         |
| <code>operatingHours</code>             | Mixed Object | Optional   Default: () =><br>cloneValue(DEFAULT_OPERATING_HOURS.Monday)    |
| <code>Tuesday</code>                    | Mixed Object | Optional   Default: () =><br>cloneValue(DEFAULT_OPERATING_HOURS.Tuesday)   |
| <code>Wednesday</code>                  | Mixed Object | Optional   Default: () =><br>cloneValue(DEFAULT_OPERATING_HOURS.Wednesday) |
| <code>Thursday</code>                   | Mixed Object | Optional   Default: () =><br>cloneValue(DEFAULT_OPERATING_HOURS.Thursday)  |
| <code>Friday</code>                     | Mixed Object | Optional   Default: () =><br>cloneValue(DEFAULT_OPERATING_HOURS.Friday)    |
| <code>Saturday</code>                   | Mixed Object | Optional   Default: null                                                   |
| <code>Sunday</code>                     | Mixed Object | Optional   Default: null                                                   |
| <code>isManuallyOpen</code>             | Boolean      | Optional   Default: null                                                   |
| <code>shopMode</code>                   | String       | Optional   Default: 'manual_open'                                          |
| <code>starvationThresholdMinutes</code> | Number       | Optional   Default: 60                                                     |
| <code>starvationBoost</code>            | Number       | Optional   Default: 20                                                     |

|      |         |                             |
|------|---------|-----------------------------|
| type | String  | Base Property Configuration |
| type | String  | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | Number  | Base Property Configuration |
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| type | Number  | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | Number  | Base Property Configuration |
| type | Boolean | Base Property Configuration |
| type | String  | Base Property Configuration |

### 3. User Planned Schema: User

| Proposed Field Name     | Data Type / Constraints | Expected Behavior         |
|-------------------------|-------------------------|---------------------------|
| name                    | String                  | Required                  |
| email                   | String                  | Required                  |
| password                | String                  | Required                  |
| collegeId               | String                  | Required                  |
| department              | String                  | Required                  |
| role                    | String                  | Required                  |
| pendingRole             | String                  | Optional   Default: null  |
| pendingRoleApproval     | Boolean                 | Optional   Default: false |
| verified                | Boolean                 | Optional   Default: false |
| verificationToken       | String                  | Optional                  |
| verificationTokenExpiry | Date                    | Optional                  |
| pendingEmail            | String                  | Optional                  |
| emailChangeToken        | String                  | Optional                  |
| emailChangeTokenExpiry  | Date                    | Optional                  |
| resetPasswordToken      | String                  | Optional                  |
| resetPasswordExpiry     | Date                    | Optional                  |
| isActive                | Boolean                 | Optional   Default: true  |
| totalOrders             | Number                  | Optional   Default: 0     |
| totalSpend              | Number                  | Optional   Default: 0     |
| phone                   | String                  | Optional                  |

|                                   |              |                             |
|-----------------------------------|--------------|-----------------------------|
| <code>sessionInvalidatedAt</code> | Date         | Optional                    |
| <code>queueAssignment</code>      | String       | Optional   Default: 'All'   |
| <code>pushSubscription</code>     | Mixed Object | Optional   Default: null    |
| <code>type</code>                 | String       | Base Property Configuration |
| <code>type</code>                 | String       | Base Property Configuration |
| <code>type</code>                 | String       | Base Property Configuration |
| <code>type</code>                 | String       | Base Property Configuration |
| <code>type</code>                 | String       | Base Property Configuration |
| <code>type</code>                 | String       | Base Property Configuration |
| <code>type</code>                 | String       | Base Property Configuration |
| <code>type</code>                 | Boolean      | Base Property Configuration |
| <code>type</code>                 | Boolean      | Base Property Configuration |
| <code>type</code>                 | String       | Base Property Configuration |
| <code>type</code>                 | Date         | Base Property Configuration |
| <code>type</code>                 | String       | Base Property Configuration |
| <code>type</code>                 | String       | Base Property Configuration |
| <code>type</code>                 | Date         | Base Property Configuration |
| <code>type</code>                 | String       | Base Property Configuration |
| <code>type</code>                 | Date         | Base Property Configuration |
| <code>type</code>                 | Boolean      | Base Property Configuration |
| <code>type</code>                 | Number       | Base Property Configuration |

|            |        |                             |
|------------|--------|-----------------------------|
| sessionId  | String | Base Property Configuration |
| device     | String | Base Property Configuration |
| lastActive | Date   | Base Property Configuration |
| type       | Date   | Base Property Configuration |
| service    | String | Base Property Configuration |
| maxPages   | Number | Base Property Configuration |
| validUntil | Date   | Base Property Configuration |
| type       | String | Base Property Configuration |

### 3.Wallet Planned Schema: wallet

| Proposed Field Name | Data Type / Constraints | Expected Behavior           |
|---------------------|-------------------------|-----------------------------|
| userId              | ObjectId (Foreign Key)  | Required                    |
| balance             | Number                  | Optional   Default: 0       |
| hasSeenOnboarding   | Boolean                 | Optional   Default: false   |
| type                | Number                  | Base Property Configuration |
| type                | Boolean                 | Base Property Configuration |

### 3.WalletTransaction Planned Schema: `WalletTransaction`

| Proposed Field Name          | Data Type / Constraints | Expected Behavior             |
|------------------------------|-------------------------|-------------------------------|
| <code>walletId</code>        | ObjectId (Foreign Key)  | Required                      |
| <code>type</code>            | String                  | Required                      |
| <code>amount</code>          | Number                  | Required                      |
| <code>orderId</code>         | ObjectId (Foreign Key)  | Optional   Default: null      |
| <code>razorpayOrderId</code> | String                  | Optional   Default: null      |
| <code>description</code>     | String                  | Required                      |
| <code>timestamp</code>       | Date                    | Optional   Default: Date.now  |
| <code>status</code>          | String                  | Optional   Default: 'success' |
| <code>type</code>            | String                  | Base Property Configuration   |
| <code>type</code>            | Number                  | Base Property Configuration   |
| <code>type</code>            | String                  | Base Property Configuration   |
| <code>type</code>            | String                  | Base Property Configuration   |
| <code>type</code>            | Date                    | Base Property Configuration   |
| <code>default</code>         | Date                    | Base Property Configuration   |
| <code>type</code>            | String                  | Base Property Configuration   |

## 4. UI/UX & Frontend Design Specifications

The frontend presentation layer will be constructed using **React.js** and **Tailwind CSS**, utilizing mobile-first responsive design principles, glassmorphism aesthetics, and comprehensive Dark Mode support.

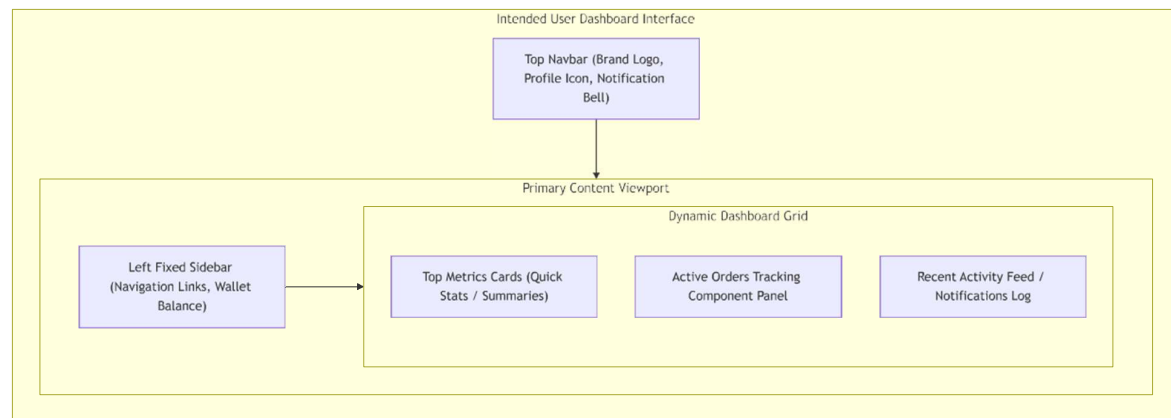
### 4.1 Proposed Design System & Guidelines

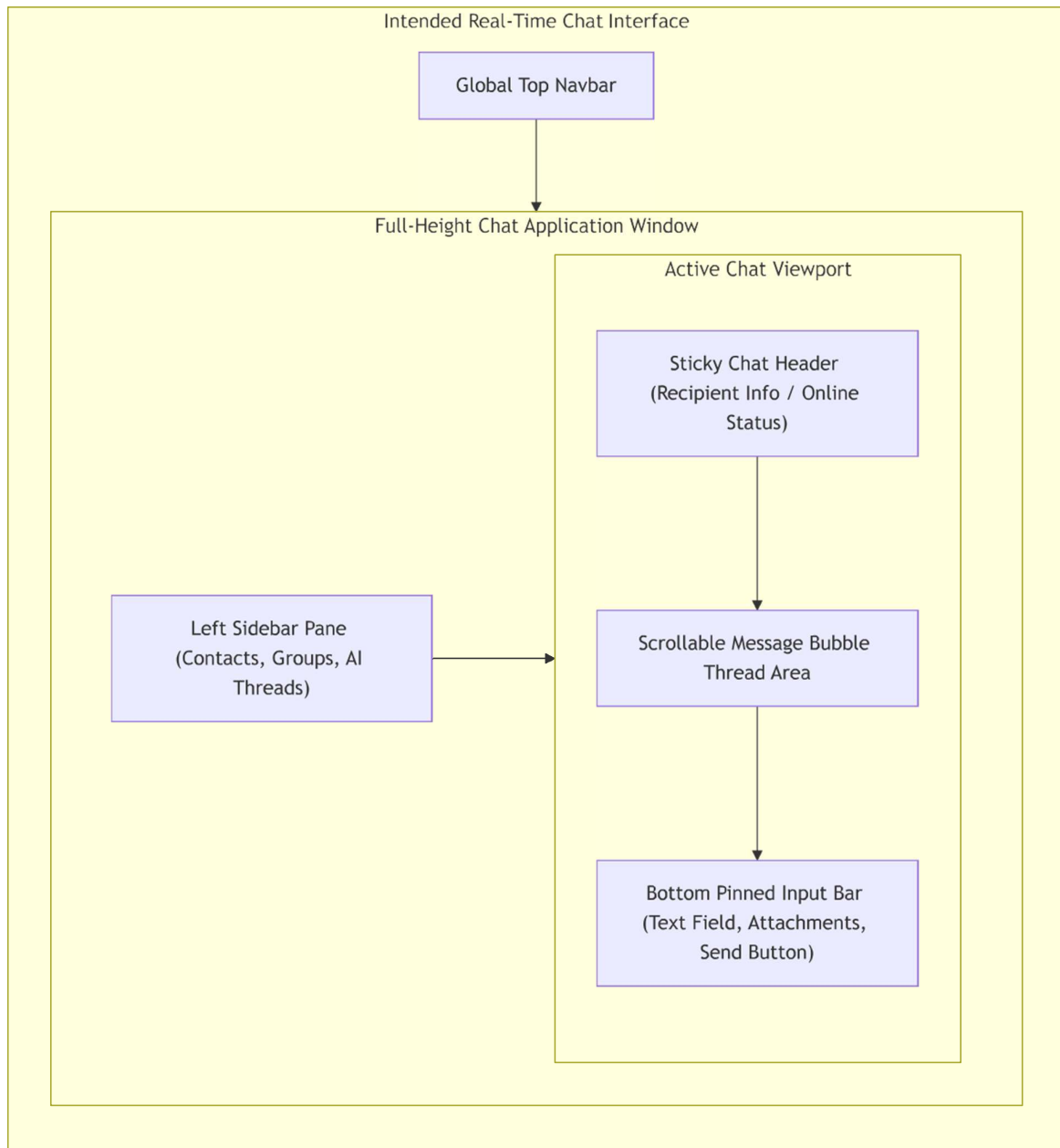
- **Typography:** Core fonts will include 'Inter' for UI elements and 'Roboto' for large text bodies, ensuring high legibility across device form factors.
- **Color Palette:** Deep Blues will be used for primary interactions, Emerald Greens for successful operations, and Rose Red for destructive or error-state actions. The dark mode theme will utilize deep slate gradients to significantly reduce eye strain.
- **Component Library:** We will construct custom-built Tailwind components paired directly with Lucide React to maintain consistent SVG iconography across the portal.

### 4.2 Major Layout Diagrams (Structural Wireframes)

Below are the structural wireframe blueprints depicting how the primary page layouts will be divided and nested within the Document Object Model (DOM).

Proposed Dashboard Layout Structure





| Planned Frontend Component / Page Name | Primary Intended Purpose                                                                                                              |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| About                                  | Will render the core About context. Needs to be integrated into the global router and tested across all mobile breakpoints            |
| ChatPage                               | Will render the core ChatPage context. Needs to be integrated into the global router and tested across all mobile breakpoints         |
| ContactUs                              | Will render the core ContactUs context. Needs to be integrated into the global router and tested across all mobile breakpoints        |
| CookiePolicy                           | Will render the core CookiePolicy context. Needs to be integrated into the global router and tested across all mobile breakpoints     |
| DocumentLab                            | Will render the core DocumentLab context. Needs to be integrated into the global router and tested across all mobile breakpoints      |
| FriendsPage                            | Will render the core FriendsPage context. Needs to be integrated into the global router and tested across all mobile breakpoints      |
| Home                                   | Will render the core Home context. Needs to be integrated into the global router and tested across all mobile breakpoints             |
| PrivacyProtocol                        | Will render the core PrivacyProtocol context. Needs to be integrated into the global router and tested across all mobile breakpoints  |
| PublicOrderTrack                       | Will render the core PublicOrderTrack context. Needs to be integrated into the global router and tested across all mobile breakpoints |
| SplitPayPage                           | Will render the core SplitPayPage context. Needs to be integrated into the global router and tested across all mobile breakpoints     |
| TechnicalHelp                          | Will render the core TechnicalHelp context. Needs to be integrated into the global router and tested across all mobile breakpoints    |
| TermsOfService                         | Will render the core TermsOfService context. Needs to be integrated into the global router and tested across all mobile breakpoints   |
| Track                                  | Will render the core Track context. Needs to be integrated into the global router and tested across all mobile breakpoints            |
| WalletPage                             | Will render the core WalletPage context. Needs to be integrated into the global router and tested across all mobile breakpoints       |
| WalletTopup                            | Will render the core WalletTopup context. Needs to be integrated into the global router and tested across all mobile breakpoints      |

## 5. Proposed Testing & Quality Assurance Plan

Software quality will be maintained through rigorous testing methodologies designed into the CI/CD pipeline right from the inception of development.

**Unit Testing:** Backend controllers and critical utility functions (such as wallet deduction logic and mathematics for bill splitting) will be unit tested using Jest/Mocha to identify failures in edge cases (e.g., negative inputs, zero values).

**Integration Testing:** All API endpoints will be continuously validated using tools like Postman to ensure that the request-response cycles match the API specifications and properly persist data into MongoDB.

**End-to-End (E2E) Testing:** Once the UI is built, simulated user workflows (e.g., Login → Upload Document → Top-up Wallet → Pay → Track) will be automated to guarantee the integrity of the user journey.

## 6. Proposed Security Architecture

Security will serve as a fundamental pillar of the Reposys infrastructure, implementing multiple layers of defense to protect user data and financial logs.

- **Data in Transit:** 100% of data payloads will be encrypted via TLS/SSL, strictly enforcing HTTPS across all public and internal endpoints.
- **Data at Rest:** Passwords will be cryptographically hashed using bcrypt with dynamic salt rounds before touching the database. Environment variables will be used exclusively to store third-party API secrets.
- **XSS Protection:** The React framework will inherently sanitize inputs and escape string injections prior to DOM rendering, preventing script executions.
- **Rate Limiting:** A robust Rate Limiting middleware will be designed to prevent DDoS attacks and brute-force login attempts on critical authentication routes.

## 7. Conclusion & Future Scope

### 7.1 Design Conclusion

The proposed Reposys project architecture successfully conceptualizes a comprehensive digital ecosystem required for modern educational campuses. By tightly defining the schemas, API routes, and interaction models for financial capabilities (Digital Wallet), operational services (Document Printing), and robust communication channels (Real-time Chat, AI Assistant), this design sets a clear and secure path for the development phase.

## 7.2 Identified System Limitations

Even before development begins, the architecture highlights the following limitations that stakeholders must accept:

- **Network Dependency:** The web-based system will rely heavily on continuous internet connectivity; offline capabilities via Service Workers will be restricted solely to static asset caching.
- **File Size Constraints:** Document uploads for printing will need to be intentionally capped (e.g., 5-10MB limits) to prevent memory exhaustion and buffer overflow on the Node.js application server.

## 7.3 Future Scope Post-Deployment

Once the initial architecture defined in this document is realized and deployed, future versions of the system may explore the following avenues:

- **Native Mobile Application:** Migrating the React web application footprint to React Native to deliver dedicated iOS and Android applications with integrated biometric authentication.
- **Advanced LLM Grading:** Extending the AI capabilities beyond user support to incorporate automated assignment grading and deep-text summarization features.